

Business Questions

1. Top Performing Products by Rating

The analysis identified the top 10 highest-rated products on Amazon. Three products achieved a perfect rating of 5.0, while the remaining products received ratings between 4.7 and 4.8. This indicates that these products consistently met customer expectations and delivered a positive user experience. Their high ratings suggest strong product quality and customer satisfaction.

2. Price vs. Rating Analysis

The comparison of actual prices, discounted prices, and customer ratings shows that there is no strong relationship between product price and customer ratings. Some products with moderate prices received ratings similar to or even higher than more expensive products. This suggests that customers are more likely to rate products based on their quality, performance, and value for money rather than price alone.

3. Category Performance

The category performance analysis shows that customer ratings remain relatively consistent across most product categories, with average ratings ranging between 4.4 and 4.6. However, customer engagement varies considerably, as some categories attract significantly more reviews than others. Categories with both high ratings and a large number of reviews demonstrate stronger customer acceptance and demand.

4. Product Popularity by User Engagement

Since each product in the dataset is associated with a single user record, Rating Count was used to measure customer engagement instead of unique User IDs. The analysis revealed that products such as the Amazon Basics High-Speed HDMI Cable, boAt BassHeads Earphones, and Redmi 9A Smartphone received the highest number of customer reviews. These products have attracted the greatest customer attention and are the most popular products in the dataset.

5. Additional Insights

Insight 1: Average Discount Percentage by Product Category

The analysis shows that Electronics and Accessories categories offer the highest average discounts, with discount percentages reaching approximately 90% in some categories. This suggests that Amazon frequently uses discounts in these categories to encourage sales and remain competitive.

Insight 2: Top 10 Most Discounted Products

The dashboard identified the products with the highest discount percentages, with the most discounted product receiving a discount of approximately 94%. Most of the remaining products in the top 10 offered discounts of around 88–91%, making them attractive options for customers looking for significant savings.